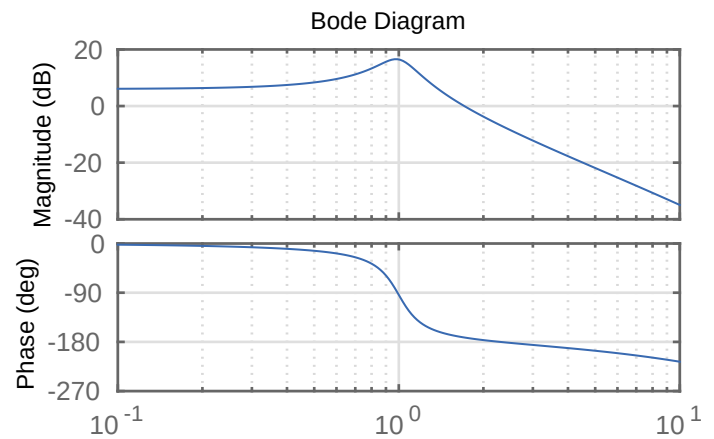


1. What is the crossover frequency, ω_c , of this system?

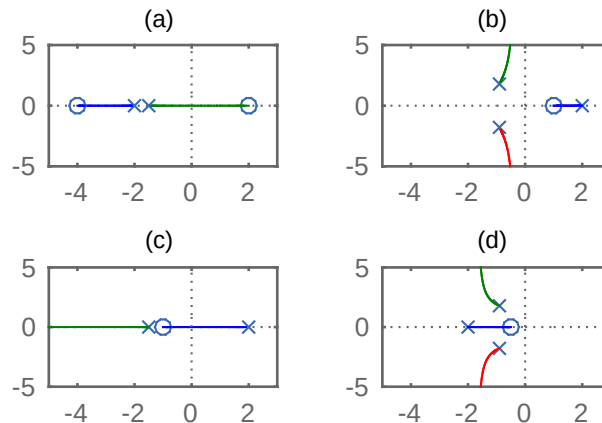
- (a) 1.5 rad/sec
- (b) 3 rad/sec
- (c) 0.5 rad/sec
- (d) 10 rad/sec



2. What is the I-part in an ideal PID controller?

- (a) $K_I s$
- (b) K_I
- (c) $\frac{K_I}{s}$
- (d) $K_I + s$

3. Which system is stable for all K?



4. How many poles does the following transfer function have?

$$G(s) = \frac{s - 8}{s^2 - 4s - 1}$$

- (a) 8
- (b) 3
- (c) 1
- (d) 2

5. Which one is the correct Taylor expansion of $\sqrt{h}$? ("h.o.t." represents high order terms)

- (a) $\sqrt{h} = \sqrt{h_0} + (h - h_0) \frac{1}{2\sqrt{h_0}} + \text{h.o.t.}$
- (b) $\sqrt{h} = \sqrt{h_0} + (h - h_0) \frac{1}{h_0} + \text{h.o.t.}$
- (c) $\sqrt{h} = 2\sqrt{h_0} + (h - h_0) \frac{1}{\sqrt{h_0}} + \text{h.o.t.}$
- (d) $\sqrt{h} = 2\sqrt{h_0} + (h - h_0) \frac{2}{h_0} + \text{h.o.t.}$

Answers:

1.	
2.	
3.	
4.	
5.	